Quantitative Finance Spring 2025 Portfolio Analysis Report

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Primary Objective:
Choosing 7 traded assets from the NYSE and with the use of pandas and seaborn libraries, compare these stocks to Dow Jones, S&P 500, and Russell 2000. Creating charts and tables to see how these stocks may correlate with one another.

Part 1 (create a table of calculated factors for each asset in the portfolio):
The seven assets that were chosen were Nike, Disney, JP morgan, Coca-Cola, Spotify, Toyota Motor corp, and AT&T. The first step in creating the table was to signify all tickers in column 1. Ensuring that the data shown is only adj close prices and that it is using data only throughout ten years.

The portfolio weight was the first calculated factor and that was found through dividing 1 by the number of tickers found in the portfolio. This ensures that all tickers are equally weighted and that there are no discrepancies.

Next was to find the annualized volatility of each asset using a trailing of three months. The explanation to the code was found through a post on stack overflow which described using the .rolling function of 13 weeks to get the rolling volatility[1]. Then using the rolling volatility to calculate the annualized volatility.

```
rolling_vol = asset_returns.rolling(91).std()
volatility = (rolling_vol.mean() * np.sqrt(252))
```

The next step in the process was to find the betas against all three indexes, SPY, DIA, and IWM. The formula and how to calculate the betas was found on a post in stack overflow describing the betas as the covariance of the stock returns and the market returns divided all by the variance of the market returns[2]. With the formula know the following code can then be created:

```
beta_DIA = asset_returns.cov(returns['DIA']) / returns['DIA'].var()
beta_SPY = asset_returns.cov(returns['SPY']) / returns['SPY'].var()
beta_IWM = asset_returns.cov(returns['IWM']) / returns['IWM'].var()
```

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[1] stackoverflow.com *https://stackoverflow.com/questions/52941128/how-to-calculate-volatility-with-pandas*
[2] stackoverflow.com
https://stackoverflow.com/questions/59110902/is-there-anyway-to-calculate-market-beta-from-yahoo-finance-datasreader-on-pytho\n

Next columns now correspond to the average weekly drawdown, and the max weekly drawdown. First I found the rolling adj close to weekly high and weekly low throughout the 52 weeks. Then Using the formula provided the following code can be made:

```
weekly_high = prices.rolling(364).max()
weekly_low = prices.rolling(364).min()\
avg_drawdown = ((weekly_low - weekly_high) / weekly_high).mean()
max_drawdown = ((weekly_low - weekly_high) / weekly_high).min()
```

For the total return formula it is the percentage change of the ending price and the starting price. It is expressed as (ending price /starting price) - 1. This nets us the total return for each asset with all the ten years of data. Then finally annualized total return which is expressed as the formula (1+total return)^1/years - 1. This allows us to understand the average compound growth rate of each asset's total return over the ten years.

```
total_return = (prices.iloc[-1] / prices.iloc[0]) - 1
annualized_return = (1 + total_return) ** (1 / 10) - 1
```

Now lastly all the data was indexed into a variable that saved the results and a dataframe was created ensuring that all columns correspond to the right values and assets.

	Ticker	Portfolio Weight	Annualized Volatility	Beta (SPY)	Beta (IWM)	Beta (DIA)	Avg Weekly Drawdown	Max Weekly Drawdown	Total Return	Annualized Return
0	NKE	0.142857	0.314263	1.044786	0.740441	1.052746	-0.389655	-0.642635	0.805394	0.060858
1	SPOT	0.142857	0.458696	1.129150	0.805873	0.891089	-0.636314	-0.783131	NaN	NaN
2	T	0.142857	0.240483	0.614864	0.442754	0.703117	-0.293939	-0.519628	1.161269	0.080117
3	JPM	0.142857	0.268254	1.075844	0.832334	1.208666	-0.395113	-0.544850	4.438541	0.184536
4	DIS	0.142857	0.296934	1.014793	0.761098	1.052731	-0.365129	-0.575256	0.143639	0.013512
5	TM	0.142857	0.228427	0.684340	0.499827	0.687325	-0.305733	-0.483796	0.516873	0.042545
6	KO	0.142857	0.180275	0.588960	0.358816	0.654999	-0.241230	-0.372624	1.319685	0.087785

Part 2 (Create a table comparing the portfolio against the three etfs)
This table would include the etf tickers and the comparisons between the other seven tickers chosen. To start the first column would include the correlation against the etf. This is found through getting the portfolio returns through the portfolio tickeres and correlating that against the etf returns with the .corr function. The next column is for the covariance which would be similarly formatted using the .cov function.

```
correlation = portfolio_returns.corr(etf_returns)
covariance = portfolio_returns.cov(etf_returns)
```

Next column requires that the tracking error is calculated. The code for the tracking error was cited from Vincent D., Portfolio Geek[3] and is as follows:

[3] Vincent D., Portfolio Geek https://portfolio-geek.com/posts/articles/tutorials/tracking-error

```
tracking_error = np.std(portfolio_returns - etf_returns) * np.sqrt(252)
```

The sharpe ratio formula uses the risk free ratio which was acquired from the 10 year US treasury bond rate[4].

```
sharpe_ratio = (portfolio_returns.mean() - risk_free_rate) / portfolio_vol
```

Finally the annualized volatility and the spread was found by solving for the etf volatility using the formula of standard deviations of the returns multiplied by the square root of 252 standing for the 252 operating days of the year.The spread is found by subtracting the two volatilities.

```
etf_vol = etf_returns.std() * np.sqrt(252)
vol_spread = portfolio_vol - etf_vol
```

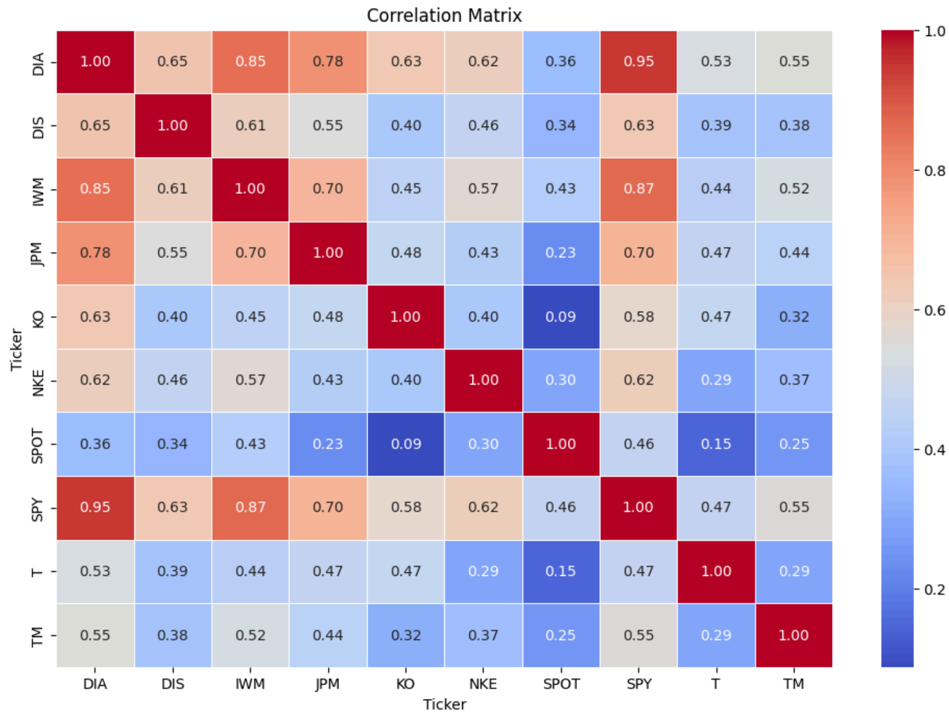
With all the columns created and data indexed, the data frame can be created and outputted as follows:

	ETF Ticker	Correlation	Covariance	Tracking Error	Sharpe Ratio	Volatility Spread
0	DIA	0.857943	0.000129	0.104027	0.001908	0.007787
1	SPY	0.854924	0.000130	0.105616	0.001908	0.005432
2	IWM	0.794845	0.000156	0.150943	0.001908	-0.050209

Part 3 (create a correlation matrix with all seven assets, etfs, and seven individual stocks):

Using the correlation function on all the returns from all the assets, a correlation matrix can be created and formatted.

[4] ychart.com https://ycharts.com/indicators/10_year_treasury_rate



Conclusion and key takeaways:

On average the Dow Jones and S & P 500 tickers have the most correlation against all of the portfolio tickers. 5 tickers not including Toyota or AT&T have a correlation of .60 or greater against DIA and SPY. This can be attributed to SPY and DIA having a correlation of .95 against each other. In comparison Russell 2000 only has 2 out of the 7 portfolio tickers that are above .60. These findings give us an understanding of how large indexes aid in the trade of assets by providing a thorough analysis of how the assets react to the grander market.